1. **Frontend (Node.js with Express)**:

- Create a frontend using Express and Node.js.
- Include a form similar to the one from the Flask Assignment 2.
- Configure the form to send a request to the Flask backend.
- **Backend (Flask)**:
- Use the Flask backend to handle the form submission and process the data.
- **Folder Structure**:
- Organize the project with separate folders for the frontend and backend.
- **Docker Configuration**:
- Create a `Dockerfile` for both the frontend and backend.
- Write a `.yaml` file (Docker Compose) to connect both services in the same network.
- Upload both images to docker hub and push your whole code to github and add the node_modules and other non required files(.vscode) in .gitignore

Link - https://github.com/Toto3107/flaskp

Submission Guidelines -: 1. Upload the terminal screen shots and all your project code or files on github.
2. Share your github repo link

GITHUB REPO LINK : https://github.com/UmeshChandra2001/DockerP1

HERE IS MY ANSWER

Install of docker in my system
CMD : **sudo apt install docker.io**

Install of docker-compose in my system
CMD : **sudo apt install docker-compose**

Validation of version
CMD : **docker --version**
OUTPUT : Docker version 28.2.2, build 28.2.2-0ubuntu1~22.04.1

CMD : **docker-compose --version**
OUTPUT : docker-compose version 1.29.2, build unknown

Project folder Structure:

```
DockerP1/
│
├── frontend/              # Node.js + Express frontend
│   ├── Dockerfile
│   ├── package.json
│   ├── server.js
│   └── views/
│       └── form.html
│
├── backend/               # Flask backend
│   ├── Dockerfile
│   ├── app.py
│   └── requirements.txt
│
├── docker-compose.yaml
└── .gitignore
```

ADD OF CODE IN VS CODE:

Frontend

```
└── views/
    └── form.html
```

CODE:

```
<!DOCTYPE html>
<html>
<head>
  <title>Express Form</title>
</head>
<body>
  <h1>Submit Data</h1>
  <form action="/submit" method="POST">
    <label for="name">Name:</label>
    <input type="text" name="name" required>
    <br><br>
    <label for="age">Age:</label>
    <input type="number" name="age" required>
    <br><br>
    <button type="submit">Submit</button>
  </form>
</body>
</html>
```

server.js

```
const express = require('express');
const bodyParser = require('body-parser');
const path = require('path');
const axios = require('axios');

const app = express();
app.use(bodyParser.urlencoded({ extended: true }));

// Serve static HTML form
app.get('/', (req, res) => {
  res.sendFile(path.join(__dirname, 'views', 'form.html'));
});

// Handle form submission and forward to Flask backend
app.post('/submit', async (req, res) => {
  try {
    const response = await axios.post('http://backend:5000/process', req.body);
    res.send(`<h2>Response from Flask Backend:</h2><p>${response.data}</p>`);
  } catch (error) {
    res.status(500).send('Error communicating with backend');
  }
});

app.listen(3000, () => {
  console.log('Frontend running on http://localhost:3000');
});
```

Package.json

```
{
  "name": "frontend",
  "version": "1.0.0",
  "description": "Express frontend for Flask backend integration",
  "main": "server.js",
  "scripts": {
    "start": "node server.js"
  },
  "author": "Your Name",
  "license": "MIT",
  "dependencies": {
    "axios": "^1.6.7",
    "body-parser": "^1.20.2",
    "express": "^4.18.2"
  }
}
```

Backend

app.py

```
from flask import Flask, request

app = Flask(__name__)

@app.route('/process', methods=['POST'])
def process():
    name = request.form.get('name')
    age = request.form.get('age')
    return f"Hello {name}, you are {age} years old!"

if __name__ == '__main__':
    app.run(host='0.0.0.0', port=5000)
```

requirements.txt

```
flask
```

Frontend Dockerfile Code

```
FROM node:20.20.2-alpine3.23
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
EXPOSE 3000
CMD ["node", "server.js"]
```

Backend Dockerfile Code

```
FROM python:3.10-alpine3.22
WORKDIR /app
COPY requirements.txt .
RUN pip install -r requirements.txt
COPY . .

EXPOSE 5000
CMD [ "python","app.py" ]
```

DOCKER-COMPOSE FILE CODE

```
version: '3'
services:
  frontend:
    build: /home/ubuntu/TuteDude-Assignments/DockerP1/Frontend
    ports:
      - "3000:3000"
    depends_on:
      - backend

  backend:
    build: /home/ubuntu/TuteDude-Assignments/DockerP1/Backend
    ports:
      - "5000:5000"
```

Docker Commands used to login

CMD : **sudo docker login -u <usernameprovided>**

Checking where logged or not

CMD : **sudo docker info**

BUILING THE DOCKER

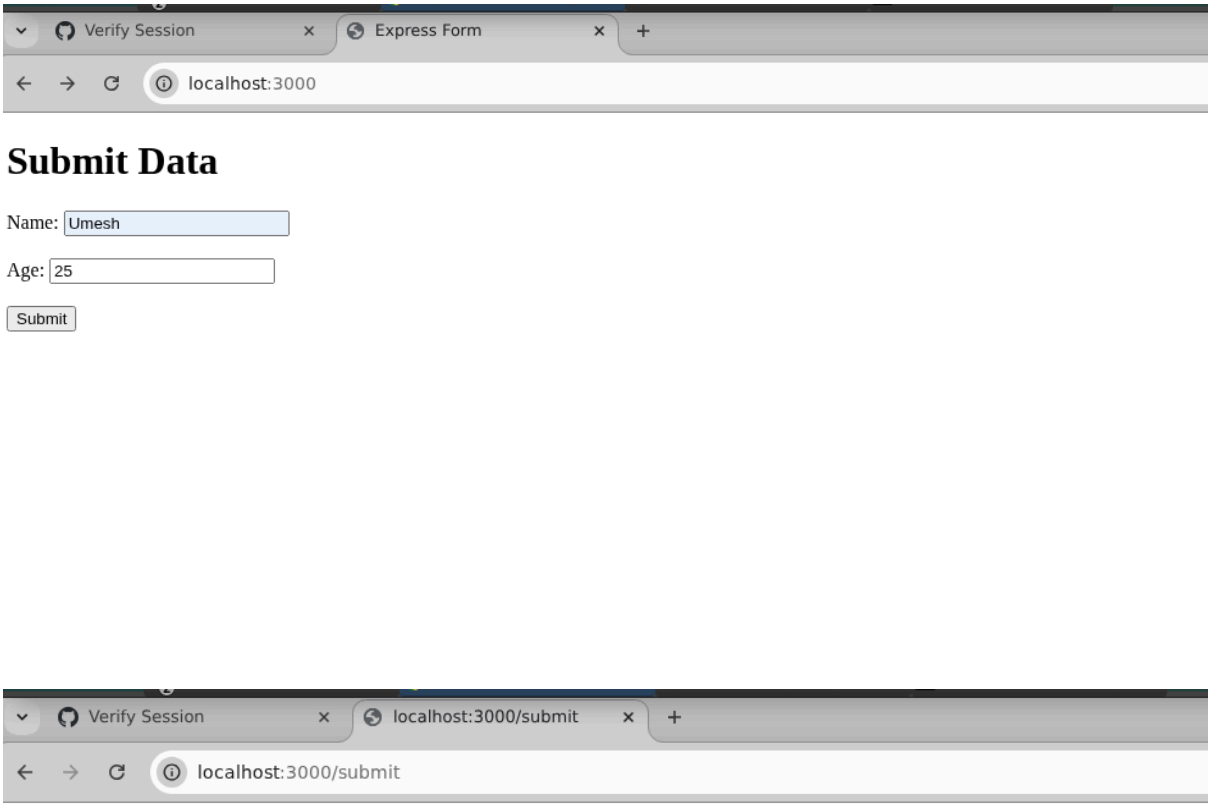
CMD : **sudo docker-compose up --build**

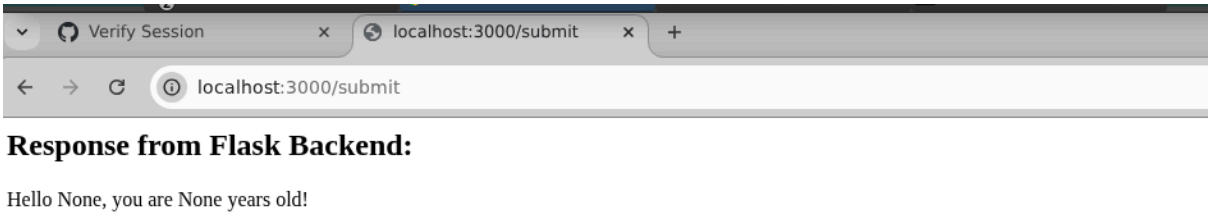
OUTPUT:

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$ sudo docker-compose up --build
Creating network "dockerp1_default" with the default driver
Building backend
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  4.096kB
Step 1/7 : FROM python:3.10-alpine3.22
3.10-alpine3.22: Pulling from library/python
d49a2dee86fb: Pulling fs layer
05a5d10d7b9a: Pulling fs layer
1a6dbb9302d8: Pulling fs layer
8b4aa26ec2ea: Pulling fs layer
8b4aa26ec2ea: Waiting
d49a2dee86fb: Verifying Checksum
d49a2dee86fb: Download complete
05a5d10d7b9a: Verifying Checksum
05a5d10d7b9a: Download complete
1a6dbb9302d8: Verifying Checksum
1a6dbb9302d8: Download complete
d49a2dee86fb: Pull complete
05a5d10d7b9a: Pull complete
1a6dbb9302d8: Pull complete
8b4aa26ec2ea: Download complete
8b4aa26ec2ea: Pull complete
Digest: sha256:27f5454a4887bb36ece39f52378a03d68ee3e1f232f90f0814fed90227a26f97
Status: Downloaded newer image for python:3.10-alpine3.22
 ---> 4857242746b8
Step 2/7 : WORKDIR /app
 ---> Running in bbd290f645f4
 ---> Removed intermediate container bbd290f645f4
 ---> 75dee737bf0c
Step 3/7 : COPY requirements.txt .
 ---> 016e0e71ce96
Step 4/7 : RUN pip install -r requirements.txt
 ---> Running in 5e3814c5ea35
Collecting flask
  Downloading flask-3.1.3-py3-none-any.whl (103 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 103.4/103.4 kB 10.2 MB/s eta 0:00:00
Collecting blinker>=1.9.0
  Downloading blinker-1.9.0-py3-none-any.whl (8.5 kB)
Collecting werkzeug>=3.1.0
  Downloading werkzeug-3.1.8-py3-none-any.whl (226 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 226.5/226.5 kB 29.8 MB/s eta 0:00:00
Collecting markupsafe>=2.1.1
  Downloading markupsafe-3.0.3-cp310-cp310-musllinux_1_2_x86_64.whl (20 kB)
Collecting itsdangerous>=2.2.0
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
```

```
Collecting markupsafe>=2.1.1
  Downloading markupsafe-3.0.3-cp310-cp310-musllinux_1_2_x86_64.whl (20 kB)
Collecting itsdangerous>=2.2.0
  Downloading itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Collecting click>=8.1.3
  Downloading click-8.3.2-py3-none-any.whl (108 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 108.4/108.4 kB 14.9 MB/s eta 0:00:00
Collecting jinja2>=3.1.2
  Downloading jinja2-3.1.6-py3-none-any.whl (134 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 134.9/134.9 kB 21.0 MB/s eta 0:00:00
Installing collected packages: markupsafe, itsdangerous, click, blinker, werkzeug, jinja2, flask
Successfully installed blinker-1.9.0 click-8.3.2 flask-3.1.3 itsdangerous-2.2.0 jinja2-3.1.6 markupsafe-3.0.3 werkzeug-3.1.8
WARNING: Running pip as the 'root' user can result in broken permissions and conflicting behaviour with the system package manager. It is recommended to use a virtual environment inste
tps://pip.pypa.io/warnings/venv

[notice] A new release of pip is available: 23.0.1 -> 26.0.1
[notice] To update, run: pip install --upgrade pip
 ---> Removed intermediate container 5e3814c5ea35
 ---> a980bc73aef1
Step 5/7 : COPY . .
 ---> 804621aca1f2
Step 6/7 : EXPOSE 5000
 ---> Running in 6cbdd023e0ba
 ---> Removed intermediate container 6cbdd023e0ba
 ---> 3c8ce7c5a41e
Step 7/7 : CMD [ "python","app.py" ]
 ---> Running in 12acbbaeba0c
 ---> Removed intermediate container 12acbbaeba0c
 ---> e399f8a0e484
Successfully built e399f8a0e484
Successfully tagged dockerp1_backend:latest
Building frontend
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  6.144kB
Step 1/7 : FROM node:20.20.2-alpine3.23
20.20.2-alpine3.23: Pulling from library/node
589002ba0eae: Pulling fs layer
d7f6aca05b78: Pulling fs layer
62e0e1181d88: Pulling fs layer
c7734204880f: Pulling fs layer
c7734204880f: Waiting
589002ba0eae: Verifying Checksum
589002ba0eae: Download complete
62e0e1181d88: Verifying Checksum
62e0e1181d88: Download complete
589002ba0eae: Pull complete
d7f6aca05b78: Verifying Checksum
d7f6aca05b78: Download complete
c7734204880f: Verifying Checksum
```

```
d7f6aca05b78: Download complete
c7734204880f: Verifying Checksum
c7734204880f: Download complete
d7f6aca05b78: Pull complete
62e0e1181d88: Pull complete
c7734204880f: Pull complete
Digest: sha256:f598378b5240225e6beab68fa9f356db1fb8efe55173e6d4d8153113bb8f333c
Status: Downloaded newer image for node:20.20.2-alpine3.23
 ---> ccfb9f0dc7ac
Step 2/7 : WORKDIR /app
 ---> Running in 6e15d5150425
 ---> Removed intermediate container 6e15d5150425
 ---> 61534f91b4f3
Step 3/7 : COPY package*.json ./
 ---> 62b26ff83c09
Step 4/7 : RUN npm install
 ---> Running in 8b54a72fe1bb

added 77 packages, and audited 78 packages in 3s

17 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
npm notice
npm notice New major version of npm available! 10.8.2 -> 11.12.1
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.12.1
npm notice To update run: npm install -g npm@11.12.1
npm notice
 ---> Removed intermediate container 8b54a72fe1bb
 ---> 608498137e41
Step 5/7 : COPY . .
 ---> 3dc2e08e7771
Step 6/7 : EXPOSE 3000
 ---> Running in f16a9d839498
 ---> Removed intermediate container f16a9d839498
 ---> ddd9bdce3766
Step 7/7 : CMD ["node", "server.js"]
 ---> Running in eaf4e3cc7a9a
 ---> Removed intermediate container eaf4e3cc7a9a
 ---> ea0b48fecc45
Successfully built ea0b48fecc45
Successfully tagged dockerp1_frontend:latest
Creating dockerp1_backend_1 ... done
Creating dockerp1_frontend_1 ... done
Attaching to dockerp1_backend_1, dockerp1_frontend_1
backend_1    |  * Serving Flask app 'app'
backend_1    |  * Debug mode: off
backend_1    | WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
backend_1    |  * Running on all addresses (0.0.0.0)
backend_1    |  * Running on http://127.0.0.1:5000
backend_1    |  * Running on http://172.18.0.2:5000
backend_1    | Press CTRL+C to quit
```


Verify Session
Express Form
localhost:3000
Submit Data
Name: Umesh
Age: 25
Submit


Verify Session
localhost:3000/submit
localhost:3000/submit
Response from Flask Backend:
Hello None, you are None years old!

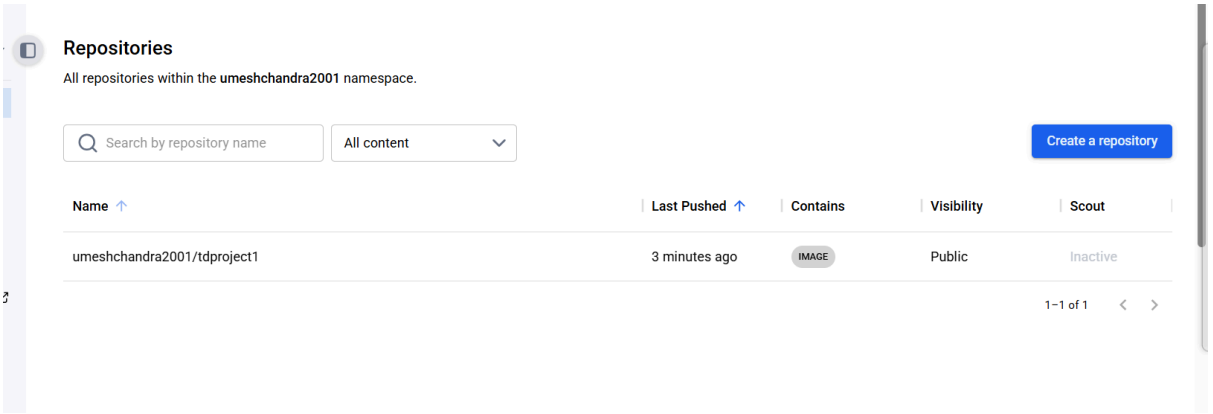
Checking images

CMD : sudo docker images

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$ sudo docker images
REPOSITORY          TAG                 IMAGE ID       CREATED          SIZE
dockerp1_frontend   latest              ea0b48fecc45   15 minutes ago   148MB
dockerp1_backend    latest              e399f8a0e484   15 minutes ago   63.3MB
node                20.20.2-alpine3.23  ccfb9f0dc7ac   9 days ago       136MB
python              3.10-alpine3.22     4857242746b8   4 weeks ago      51.4MB
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$
```

PUSHING TO DOCKER:

CREATED A REPOSITORY CALLED **tdproject1**



Tag images for Docker Hub:

CMD :sudo docker tag dockerp1_frontend:latest umeshchandra2001/tdproject1:fronten

CMD : sudo docker tag dockerp1_backend:latest umeshchandra2001/tdproject1:backend

Verify login:

CMD : sudo docker info | grep Username
Username: umeshchandra2001

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1/Backend$ sudo docker tag dockerp1_frontend:latest umeshchandra2001/tdproject1:fronten
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1/Backend$ sudo docker tag dockerp1_backend:latest umeshchandra2001/tdproject1:backend
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1/Backend$ sudo docker info | grep Username
 Username: umeshchandra2001
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1/Backend$ sudo docker push umeshchandra2001/tdproject1:frontend
```

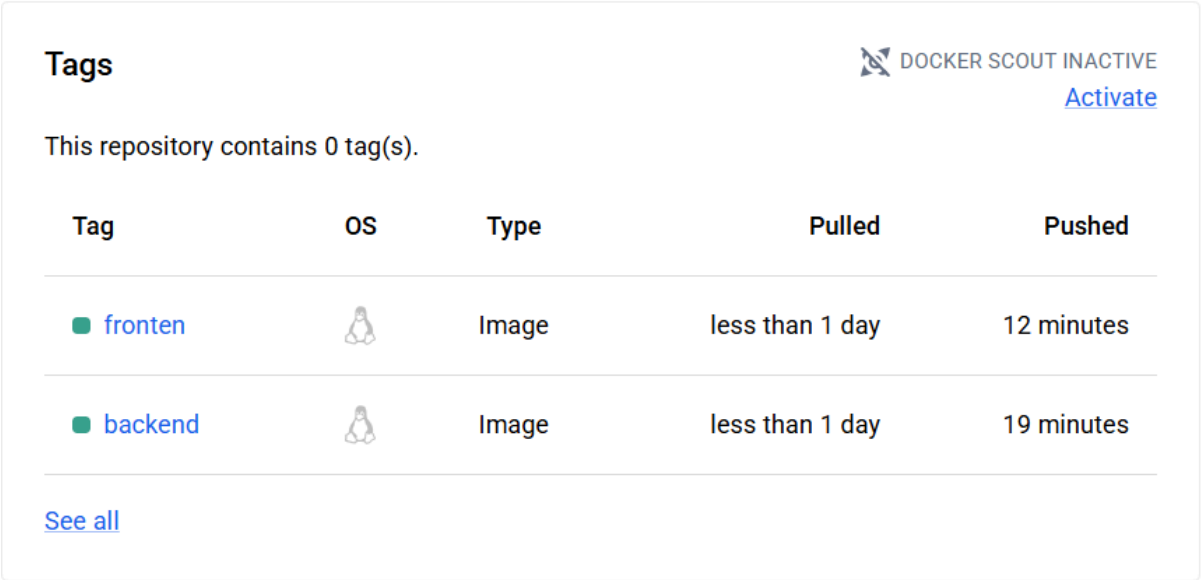
Push images

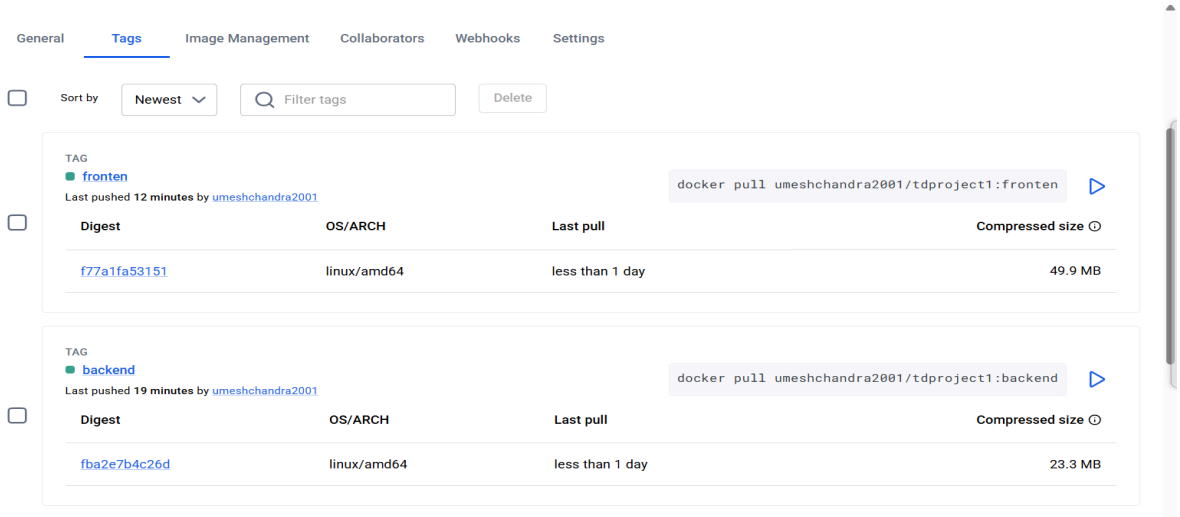
CMD : sudo docker push umeshchandra2001/tdproject1:backend

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1/Backend$ sudo docker push umeshchandra2001/tdproject1:backend
The push refers to repository [docker.io/umeshchandra2001/tdproject1]
adfb9ca0dd2b: Pushed
7f654a09b716: Pushed
e13e1dfb57c2: Pushed
013bce42f61e: Pushed
cf43eeae7c5d: Mounted from library/python
7ea0976f4b2e: Mounted from library/python
fe81b6fa373b: Mounted from library/python
cce92674e987: Mounted from library/python
backend: digest: sha256:fba2e7b4c26dd5ff46c5675b126d3b35f01594b3f2fe2cbe39ee232f414ef427 size: 1988
```

CMD : sudo docker push umeshchandra2001/tdproject1:fronten

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1/Frontend$ sudo docker push umeshchandra2001/tdproject1:fronten
The push refers to repository [docker.io/umeshchandra2001/tdproject1]
9efc39c67af5: Pushed
7ddbdbe31d47: Pushed
70eb2d53d897: Pushed
4c53c1aa63ca: Pushed
52caa5bd1d9d: Pushed
5906df279b58: Pushed
466f2827c5d1: Pushed
989e799e6349: Pushed
fronten: digest: sha256:f77a1fa531513cee6af954354f91d207ab3f11b4e76a28476fb56e83bc676e49 size: 1990
```





PUSHING TO GIT

Initialize Git in your project

CMD : git init

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/ubuntu/TuteDude-Assignments/DockerP1/.git/
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$
```

Creating .gitignore **file**

CODE :

node_modules/
.vscode/
__pycache__/
*.pyc

Stage and commit your code

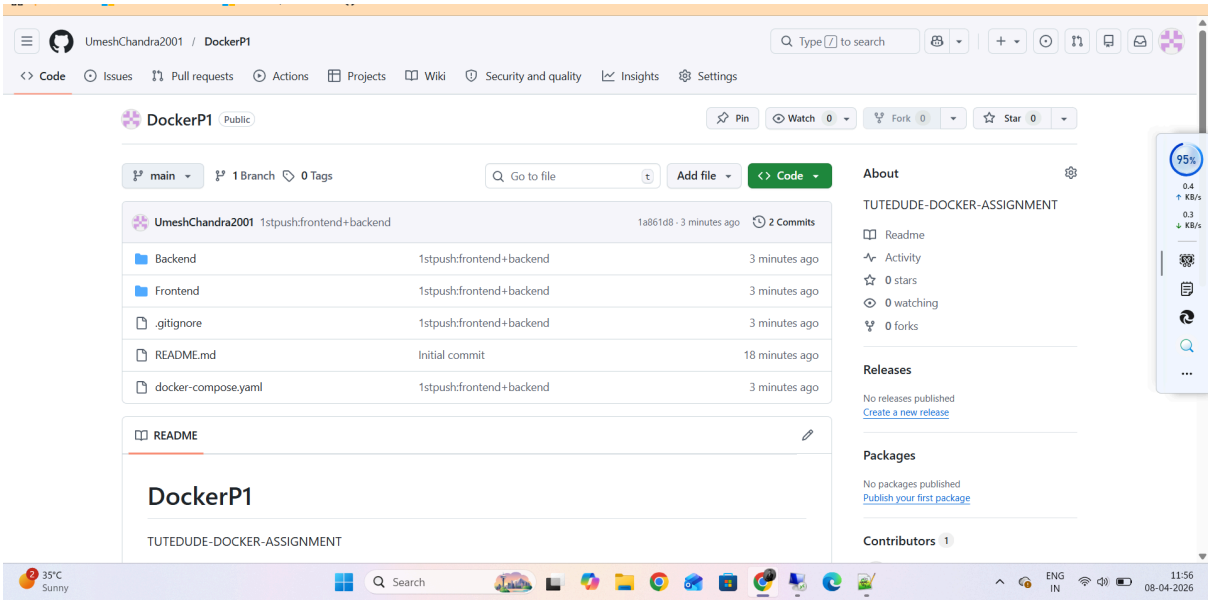
CMD : git add .
CMD : git commit -m “1stpush:frontend+backend”

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$ git add .
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$ git commit -m "1stpush:frontend+backend"
[master (root-commit) 17ec130] 1stpush:frontend+backend
 9 files changed, 106 insertions(+)
 create mode 100644 .gitignore
 create mode 100755 Backend/Dockerfile
 create mode 100755 Backend/app.py
 create mode 100755 Backend/requirements.txt
 create mode 100755 Frontend/Dockerfile
 create mode 100755 Frontend/package.json
 create mode 100755 Frontend/server.js
 create mode 100755 Frontend/views/form.html
 create mode 100755 docker-compose.yaml
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$
```

CONNECTING GIT HUB

CMD : git remote add origin https://github.com/UmeshChandra2001/DockerP1.git

PUSHING TO GIT :

CMD : **git pull origin main --rebase**
CMD : **git push -u origin main**

```
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$ git pull origin main --rebase
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 906 bytes | 906.00 KiB/s, done.
From github.com:UmeshChandra2001/DockerP1
 * branch            main       -> FETCH_HEAD
 * [new branch]      main       -> origin/main
Successfully rebased and updated refs/heads/main.
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$ git push -u origin main
Enter passphrase for key '/home/ubuntu/.ssh/id_ed25519':
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 2 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (14/14), 2.24 KiB | 2.24 MiB/s, done.
Total 14 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:UmeshChandra2001/DockerP1.git
   ee2d8a6..1a861d8  main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
ubuntu@ip-172-31-16-42:~/TuteDude-Assignments/DockerP1$
```